The Cubic Pairings Tile One Way

Every Caesium Chloride Pair and Every Delian Pair Fills Space Only as Its Own Crystal



Theorem: every pair of the caesium chloride walk from the octahedron to the cube, and every pair of the Delian walk from the rhombic dodecahedron to the cube except the octet, fills space in exactly one way, face to face or not; the rhombic dodecahedron's centre pyramids, in contrast, tile face to face in uncountably many ways, one free choice of diagonal on every face of the cube lattice

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Paper #93 · version 1 · 4 November 2026 · Licence CC BY-SA 4.0 · Scripts, exact checks and figures: https://github.com/DICTOR-Master/kaleidohedra/tree/research/studies/2026-11-04-cubic-pairings (research branch)
The idea is DICTO's: to carry the method of #92 (edge lengths, face shapes and dihedral angles decide the tilings) over to the cubic pairings. Computation, proofs and drafting with Claude (Anthropic).

Status (draft; this box is deleted at publication)

- Draft for DICTO's reading, 4 November 2026. Not yet independently reviewed. DICTO DOI: to be reserved; it goes on the meta line at publication, with the commit of the study folder. No “View in app” link yet.
- Names for DICTO to choose (descriptive placeholders used here): the square bipyramid **B** of two cube pyramids across a face; the cube's centre pyramid **C6** and the rhombic dodecahedron's **R12**; the family of R12 tilings of Theorem B (“diagonal-choice tilings”).
- Prior art not read in full: Grünbaum 1994 (cited from the Wikipedia list), Kelvin 1887, Barlow 1883 (as in #92).

Abstract. Paper #90 (Proposition 6) placed the sections $P_\mu = C \cap \mu C^*$ and $Q_\mu = C \cap (3-\mu)C^*$ of a cube and its polar octahedron in caesium chloride (CsCl) pairs that fill space for every μ, from the cuboctahedron and octahedron through DICTO's Unborn Axes and Icosaxes, Kelvin's cell and the Golden Cut Cubes to the cube; #51 gave a second walk, T(t) and O(t), through the octet from Kepler's rhombic dodecahedron to the cube. **Main theorem.** (A) For every μ in (1, 3), every tiling by copies of P_μ and Q_μ, face to face or not, is the CsCl tiling; (B) for every t in $(0, 1/3) \cup (1/3, 1)$, every tiling by T(t) and O(t) is #51's. No pair on either walk has a layer variant: #92's freedom lives only at the octet, the sliding freedom only at the cube; Kelvin's cells and Kepler's rhombic dodecahedra each tile in one way only. (C) The rhombic dodecahedron's centre pyramids are free: their face-to-face tilings correspond one to one with the choices of a diagonal on every square face of the cube lattice, so #77's law that face pyramids rebuild their solid fails for the rhombic dodecahedron. (D) The cube's centre pyramids tile face to face only as cubes (#77); otherwise their lateral edges are shared end to end by three, and no finite list describes the tilings. The proofs use the angle sums round edges and one lemma on faces; 67 exact checks.

Cut a cube's corners with a growing octahedron and a second, smaller solid appears in the gaps; together the two fill space as caesium and chloride ions do in a salt crystal. We ask whether the two pieces could fill space in any other way, as the octet's pieces can be stacked in layers. They cannot: along the whole walk the pieces lock into their crystal and nowhere else, because at every edge the angles leave only one choice and no edge can rest

> *on a face. The pyramids of the rhombic dodecahedron are free instead: on every face of a cube grid, one of two diagonals can be chosen.*

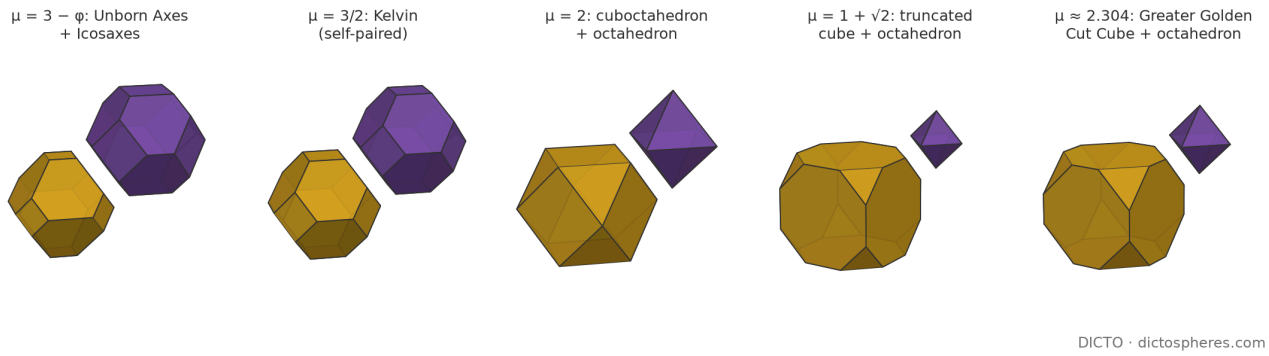


Figure 1. Five stations of the CsCl walk (#90, Proposition 6): the cell P_μ at a site of $2\mathbb{Z}^3$ (gold) and its partner Q_μ at the body centre (purple), drawn apart across their shared face. At each station, and at every μ between, this is the only way the two pieces fill space (Theorem A).

1. Setting and main theorem

The CsCl walk. Put $C = [-1, 1]^3$ and $C^* = \{|x| + |y| + |z| \le 1\}$. For $1 < \mu < 3$ let $P_\mu = C \cap \mu C^*$ at the points of $2\mathbb{Z}^3$ and $Q_\mu = C \cap (3 - \mu)C^* + (1, 1, 1)$ at the points of $2\mathbb{Z}^3 + (1, 1, 1)$; these cells fill space, the *CsCl tiling* (#90, Proposition 6 [1]; the weight walk of #52 and #53 [2, 3]; #48's T(x) [4]). For $1 < \mu < 2$ both are truncated octahedra; P_μ has squares of edge $a = \sqrt{2}(\mu - 1)$ and hexagons with edges a and $b = \sqrt{2}(2 - \mu)$ alternating, and $Q_\mu = P_{3-\mu}$ has the two lengths exchanged. Kelvin's cell is $\mu = 3/2$ ($P = Q$), DICTO's Unborn Axes and Icosaxes are $\mu = 3 - \varphi$ and φ [1]. At $\mu = 2$, P is the cuboctahedron and Q the octahedron. For $2 < \mu < 3$, P is a cube with cut corners (octagons and triangles; the Archimedean truncated cube at $1 + \sqrt{2}$, the Greater and Lesser Golden Cut Cubes at $\mu_G \approx 2.304$ and $\mu_L \approx 2.534$ [1]) and Q is a regular octahedron of edge $\sqrt{2}(3 - \mu)$, the edge of P's triangles. The walk is symmetric: μ and $3 - \mu$ give the same tiling with P and Q exchanged, so $\mu = 1$ is the cuboctahedron again.

The Delian walk. Paper #51 [5] cuts every double cone of the octet (from a tetrahedron's centre through a shared face to an octahedron's centre) by one plane parallel to the face, at fraction t, and gives each side to its own centre. For $0 < t < 1/3$, T(t) is a regular tetrahedron and O(t) the rhombic dodecahedron with its eight three-fold corners cut (8 triangles, 12 hexagons); for $1/3 < t < 1$, O(t) is a regular octahedron and T(t) the cube with four alternate corners cut (4 triangles, 6 hexagons), Samson and Delilah at the Delian bond. At $t = 1/3$ the pieces are the octet's solids, at $t = 0$ the rhombic dodecahedron alone, at $t = 1$ the cube.

Tilings. As in #92 [6]: congruent copies of the given pieces at the given size (mirror images allowed; all pieces here are mirror-symmetric), disjoint interiors, finitely many meeting any bounded set. Face to face means any two tiles meet in a common face of both or not at all. We write $\alpha = \arccos(1/3) \approx 70.53°$ for the tetrahedron's dihedral angle; α/π is irrational, because $2 \cos \alpha = 2/3$ is not an algebraic integer.

> **Main theorem.** (A) For every μ in (1, 3), every tiling of space by copies of P_μ and Q_μ is congruent to the CsCl tiling. In particular Kelvin's truncated octahedra tile only in the body-centred cubic way, cuboctahedra with octahedra only as the rectified cubic honeycomb, and truncated cubes (Archimedean or not, the Golden Cut Cubes among them) with their octahedra only as in the CsCl tiling.

(B) For every t in (0, 1/3) ∪ (1/3, 1), every tiling by copies of T(t) and O(t) is #51's tiling. At t = 0, Kepler's rhombic dodecahedra tile only in the face-centred cubic way.
(C) The face-to-face tilings by copies of the rhombic dodecahedron's centre pyramid R12 are exactly the tilings obtained thus: on every square face of the cube lattice $\mathbb{Z}^3$ put the square bipyramid B of the two cube pyramids across it, and cut B into two R12 by the plane through one of the square's two diagonals; the choices are independent. Allowing also the cut by the square itself, the same holds for the face-to-face tilings by R12 and the cube's pyramid C6 together. An R12 belongs to a whole rhombic dodecahedron exactly when all twelve squares round its apex are cut through that apex; the tilings made only of whole rhombic dodecahedra are the two (congruent) face-centred cubic ones.
(D) The face-to-face tilings by the cube's pyramid C6 are the cubic lattice cut into pyramids (#77). In every tiling by C6, every lateral edge is shared, end to end, by exactly three C6 and lies in no face of another tile; every tiling by unit cubes, each cut into six C6, is a tiling by C6, so no finite list of parameters describes these tilings.

> *In plain words: along both walks the pieces can fill space only as their own crystal. The freedom found in #92 belongs to the regular tetrahedron and octahedron together, and the walks pass through that point only once. The rhombic dodecahedron's pyramids are the opposite case: they fill space in more ways than can be listed, even face to face.*

2. The angles round an edge

> *The dihedral angles of every piece are made of π and the one irrational angle α, so the angle sum round an edge allows only very few combinations.*

By #44, Lemma 1 [7], at all but finitely many points x of an edge of a tile the wedges of the tiles containing x add up to 2π, or to π when x lies inside a face of another tile (the angle sum also used by Gabbrielli, Jiao and Torquato [8]). The dihedral angles are: $\beta = \pi/2 + \alpha/2 \approx 125.26°$ ($\cos\beta = -1/\sqrt{3}$) between a square or octagon and a hexagon or triangle of the CsCl cells; $\gamma = \pi - \alpha \approx 109.47°$ ($\cos\gamma = -1/3$) between two hexagons of a truncated octahedron and at the octahedron's edges; π/2 between two octagons of a truncated cube; for the Delian walk α (tetrahedron), $\pi - \alpha/2$ at the triangles and 2π/3 between hexagons of the cut rhombic dodecahedron, $\pi - \alpha$ (octahedron), $(\pi + \alpha)/2$ at the triangles and π/2 between hexagons of the cut cube. Since α/π is irrational, comparing the coefficients of α and of π in each sum gives Table 1.

Pieces	Round an edge (2π)	On a face (π)
CsCl walk, $1 < \mu \le 2$	$2\beta + \gamma$	none
CsCl walk, $2 < \mu < 3$	$2\beta + \gamma$; $4 \cdot \pi/2$	$2 \cdot \pi/2$
Delian walk, $0 < t < 1/3$	$\alpha + 2(\pi - \alpha/2)$; $3 \cdot 2\pi/3$	none
Delian walk, $1/3 < t < 1$	$(\pi - \alpha) + 2(\pi + \alpha)/2$; $4 \cdot \pi/2$	$2 \cdot \pi/2$
cube pyramid C6 (π/4 at the base, 2π/3 at the lateral edges)	$8 \cdot \pi/4$; $3 \cdot 2\pi/3$	$4 \cdot \pi/4$
RD pyramid R12 (π/3, π/2, 2π/3)	seven combinations	three combinations

Table 1. All combinations of wedges round a generic point of an edge (pairing_checks.py, Part A, exact enumeration). In each walk the γ-type angle (γ, α or $\pi - \alpha$) occurs at most once, and only the right angles of the truncated and cut cubes can lie on a face.

Call an edge *stiff* if its dihedral angle occurs in no “on a face” combination: every edge of the CsCl cells for $\mu \le 2$ and of the Delian pieces for $t < 1/3$, and every edge except the right-angled ones for $\mu > 2$ and $t > 1/3$. No segment of a stiff edge lies inside a face of another tile.

Lemma 1 (stiff faces match exactly). Let F be a face of a tile X all of whose edges are stiff. Every tile Y that meets the relative interior of F in a two-dimensional set does so in a face G ⊆ F; if all edges of G are stiff too, G = F, and Y is the only tile across F. *Proof.* Y lies on the other side of the plane of F, so it meets that plane in a face G. If a point of ∂F lay in the relative interior of G, a segment of an edge of X would lie inside a face of Y; so the connected set relint G misses ∂F, meets relint F, and lies in it: G ⊆ F. If G's edges are stiff, the same argument with X and Y exchanged gives F ⊆ G. ∎

Lemma 2 (one neighbour per face). If the face G = F of Y is fixed, and it is known which edges of F carry which dihedral angle of Y, then Y is unique: for every face of every piece here, the stabiliser of the face in the piece's symmetry group equals the group of isometries of the face that keep these edge labels (data/pairing-checks.json; for all faces and stations checked, including Kelvin's regular hexagons, where the unlabelled hexagon has 12 isometries and the stabiliser 6). *Proof.* The placements of Y with face G on F, on the far side, correspond to the labelled isometries of F modulo the stabiliser. ∎

3. Proof of (A) and (B): the pieces lock

> *Across every face the neighbour is forced, so one tile decides the whole tiling.*

The CsCl walk, 1 < μ ≤ 2. All edges are stiff, so by Lemma 1 every face of every tile is a whole face of exactly one other tile. Let F be shared by X and Y. At each edge e of F the two tiles contribute their dihedral angles, and by Table 1 the third tile Z round e completes 2β + γ. If X and Y both had β at e (F is a hexagon or triangle of both), Z would have γ, so both of Z's faces at e are hexagons (for μ = 2: triangles of an octahedron); but Z lies across X's other face at e, a square, which Lemma 1 says is a whole face of Z: impossible. Two γ at e are excluded by Table 1. So on a hexagon (or triangle) the two tiles have different angles at every edge: Y's hexagon–hexagon edges are X's square–hexagon edges. On a square both have β everywhere. In each case the labels of F for Y are forced and Lemma 2 gives Y: it is X's neighbour in the CsCl tiling (for μ ≠ 3/2 across a hexagon of P it is a Q, since its hexagon–hexagon edges have P's square edge length). Since the tiles of any tiling are joined to each other through faces, the CsCl tiling built from one tile is the whole tiling. At μ = 3/2 this excludes the mirror image of a truncated octahedron in its hexagon, the only other placement (Lemma 2), so Kelvin's cell tiles only as bcc.

The CsCl walk, 2 < μ < 3. The triangles of P and the faces of Q have only stiff edges. By Lemma 1 a face G of a tile across a triangle F lies in F; G cannot be an octagon: its four stiff edges cannot enter the relative interior of F, so they would lie on the three sides of F, although two of them are parallel; so G is a triangle, G = F. Two octahedra cannot share a face (two γ, Table 1). Two truncated cubes X, Y cannot share a triangle: at its edge e the third tile is an octahedron Z, whose triangle across X's octagon at e would have to be a whole face of X, that is, the octagon. So every triangle is shared by one P and one Q, placed by Lemma 2. From any Q the eight P on its faces, from each P the eight Q on its triangles, and so on, give the CsCl tiling, whose P–Q contacts are connected; it fills space, so it is the tiling. The right-angled edges of P (Table 1, the only ones that could lie on a face) play no part.

The Delian walk. For 0 < t < 1/3 all edges are stiff and every face is shared whole. On a triangle, two cut rhombic dodecahedra would leave a tetrahedron as third tile, lying across a hexagon: impossible; two tetrahedra would give 2α at an edge, excluded. On a hexagon both tiles are cut rhombic dodecahedra and the labels are forced (the cut edges carry π − α/2, the

others 2π/3). Lemma 2 places every neighbour. For 1/3 < t < 1 the triangles are stiff, and the argument for μ > 2 applies word for word: a hexagon of the cut cube has two parallel stiff (cut) edges and cannot lie in a triangle; triangles are shared by one octahedron and one cut cube. At t = 0, the rhombic dodecahedron alone has only the angle 2π/3 (Table 1: three per edge, none on a face), its faces are shared whole, and Lemma 2 (rhombus stabiliser 4 = isometries 4) gives the fcc tiling. ∎

Verification. pairing_checks.py builds every cell from half-spaces and checks, at fifteen stations of the CsCl walk (rational μ in Fractions; $3 - \varphi$, φ, $1 + \sqrt{2}$, μ_G, μ_L, $2 + 1/\varphi^2$, $2 + 1/\varphi$ with 60-digit arithmetic, their volumes P + Q = 8 in closed form) and nine of the Delian walk (among them both equal-edged members, the half-and-half member and the Delian bond): the face types, the edge lengths and dihedral angles (with μ symbolic), every facet of P at 0 and Q at (1, 1, 1) matched by exactly one CsCl tile of the expected kind, and the stabiliser of every face against its labelled and unlabelled isometry groups (equal except for Kelvin's hexagons and the silver octagons, where the labels decide).

4. The centre pyramids of the cube and of the rhombic dodecahedron

> *Two cube pyramids across a face make a squat double pyramid. It splits in three ways: through the square into the two cube pyramids again, or through either diagonal into two pyramids of the rhombic dodecahedron. That one choice per face is the whole freedom.*

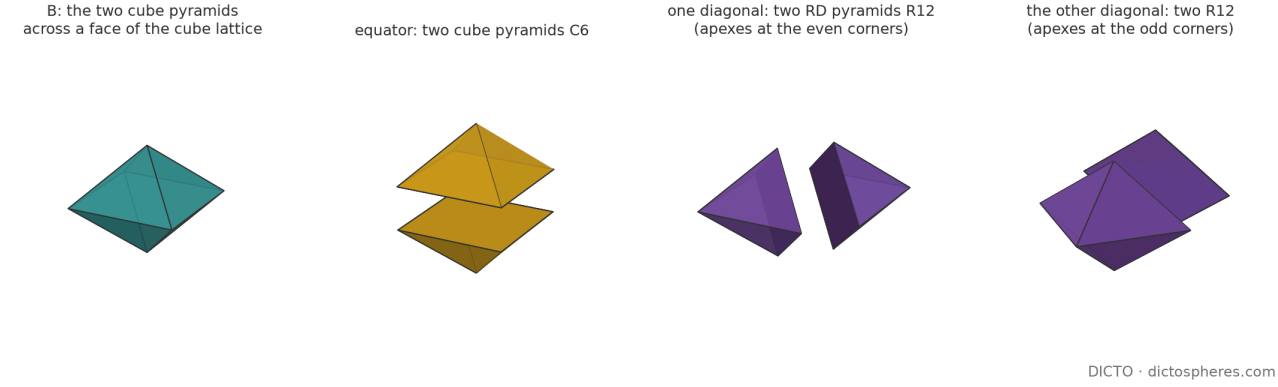


Figure 2. The bipyramid B on a face of the cube lattice $\mathbb{Z}^3$ (apexes at the two cube centres) and its three cuts: by the square into two cube pyramids C6, and by either diagonal plane into two rhombic-dodecahedron pyramids R12, with their apexes at the ends of the chosen diagonal.

The pieces. With unit cubes, C6 is the cone from a cube's centre over a face: square base of edge 1, height 1/2, lateral edges √3/2, dihedral angles π/4 at the base and 2π/3 at the lateral edges, volume 1/6. The rhombic dodecahedron {|x| + |y|, |y| + |z|, |z| + |x| ≤ 1} (the cube with six C6 added, volume 2) has twelve centre pyramids R12, each with a rhombic base of edge √3/2 (diagonals √2 and 1), lateral edges 1 (dihedral π/2) and √3/2 (dihedral 2π/3), base dihedral π/3, volume 1/6. Both have the same lateral face, the triangle (1, √3/2, √3/2); in C6 its two equal sides meet at the apex, in R12 at a three-fold corner of the rhombic dodecahedron, so in R12 the apex is not marked by the face. The bipyramid B with a unit square as equator and its apexes 1/2 above and below is the union of two C6 (cut by the square) and also of two R12 (cut by either diagonal plane through the apexes); the rhombic dodecahedron is six such B round its centre, a classical dissection [9].

Proof of (C). Let a face-to-face tiling by R12 (and C6) be given. A rhombus is a face only of R12, so the tile across it is an R12 with that rhombus; the rhombus has 4 isometries and R12 is kept by the reflections in its two mirror planes, so the tile is R12's mirror image (Lemma 2),

and the two make a B. Likewise a C6 square is matched by a C6, again a B. These B tile space face to face: the faces of the halves on the boundary of B are B's own triangles, edges and corners, so B ∩ B′ is a face of both. Cut every B through its square: this gives a face-to-face tiling by C6, which by #77, Theorem 1 [10], is the cubic lattice cut from its centres. So the B are the bipyramids over the faces of a cube lattice, and each tile is one of the halves of B in one of its three cuts (the only planes through four corners of B forming a rhombus of edge √3/2 or the square). Conversely, any choice gives a face-to-face tiling, because every triangle of B lies whole in one half and the cut adds no corners. The twelve R12 with apex at a lattice corner v are exactly the pieces of the twelve B round v cut through v, and they make the rhombic dodecahedron round v; if every tile lies in such a solid, then with v also every corner joined to v by a face diagonal is an apex, so the apexes are one of the two face-centred cubic lattices of even or of odd corners. ∎

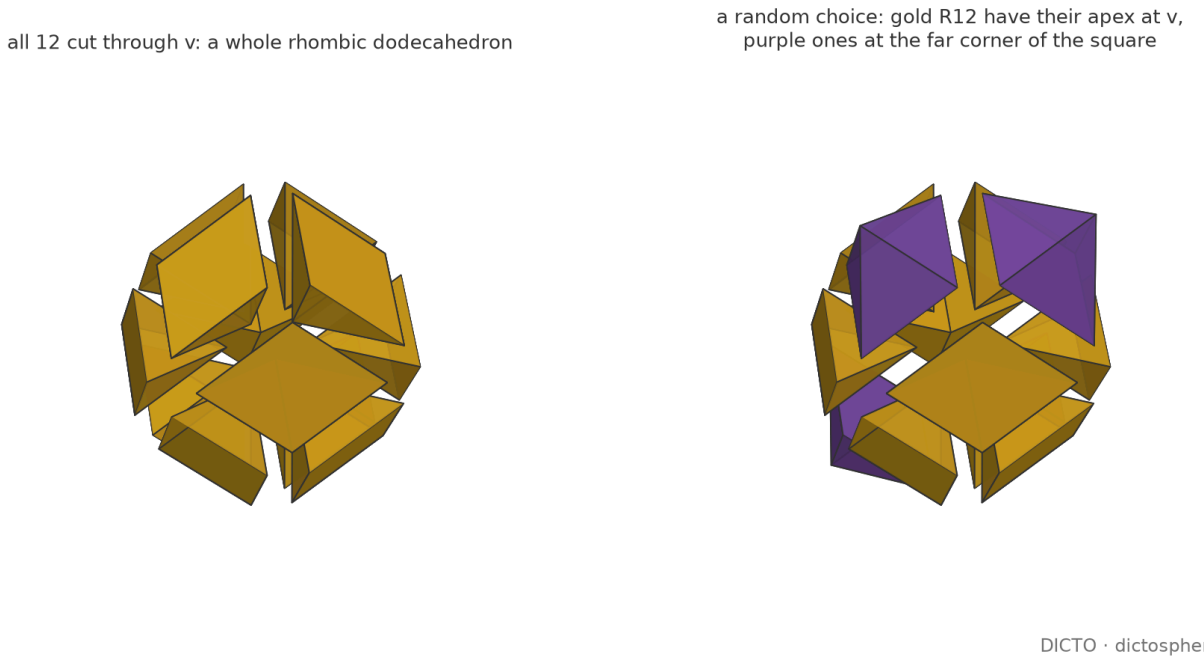


Figure 3. The twelve squares of the cube lattice at a corner v, each with the R12 its cut gives, drawn apart from v. Left: every cut through v, the whole rhombic dodecahedron. Right: a random choice; gold pyramids have their apex at v, purple ones at the opposite corner of their square, where they belong to another, incomplete rhombic dodecahedron. Every such choice, on every face, is a face-to-face tiling (Theorem C).

So the face-to-face rule does not hold the rhombic dodecahedron's pyramids together, as #77 found it does for the five Platonic solids. The difference is the lateral face: for a Platonic face pyramid its equal sides meet at the apex, which therefore passes to the neighbour; for R12 the neighbour can turn round and put its apex at the other end of the long edge. Theorem C has no layers: the hexagonal-close-packed partner of the rhombic dodecahedron, the trapezo-rhombic dodecahedron, has six trapezoidal faces, so its tiling is not one by R12, and no straight-or-twisted variant arises.

Proof of (D). Face to face: #77. In general, Table 1 shows that a lateral edge of C6 (angle 2π/3) has exactly three lateral wedges round it and never lies on a face, and a base edge has eight base wedges, or four on a face. The proof of #92, Proposition 7 [6], then applies word for word (it uses only that the lateral face's apex angle, here arccos(1/3), is less than π): the three tiles round a lateral edge e have e itself as a lateral edge, end to end. A tiling by unit cubes cut into C6 is a tiling by C6; cube tilings include the columns of the cubic lattice slid by independent amounts along one axis, so no finite list of parameters describes the tilings by

C6. Cutting each rhombic dodecahedron of the fcc tiling into a cube and six C6 gives the cubic lattice cut into C6 again. ∎

Verification. pairing_checks.py (Part D, Fractions): B = 2 C6 = 2 R12 (both diagonals); R12 is the face pyramid of the rhombic dodecahedron of volume 2; the edge lengths, the common lateral triangle and the dihedral angles of C6 and R12; three patches of 384 tiles each in the box $[-2, 2]^3$ (all R12 with random diagonals; C6 and R12 mixed at random; all C6), in which every facet of every inner tile is a whole facet of exactly one other tile and no corner of any tile lies in another inner tile, its faces or its edges; and the cut through all even corners, in which every even corner is the apex of twelve R12 (13 whole rhombic dodecahedra inside the box) and no odd corner is an apex.

5. What is settled and what stays open

Settled. All tilings by the pairs of both walks, face to face or not (Theorem A, B): each is its own crystal and nothing else. The face-to-face tilings by R12, and by R12 with C6 (Theorem C). For C6: the face-to-face case (#77), the rule for lateral edges and the absence of a finite description (Theorem D). Question 1 of the study, whether the CsCl pairs have layer variants, is answered no at every station, the golden ones included.

Open, not conjectured. (1) Whether in every tiling by C6 the pyramids fall into whole cubes, that is, whether a C6 round a lateral edge can have its apex at the far end (the open point of #92, Proposition 7, for the cube). (2) The tilings by R12 that are not face to face: Table 1 leaves seven combinations round an edge and three on a face, and we have not classified them. (3) Tilings by the pieces of the walks at different sizes, and by the trapezo-rhombic dodecahedron's centre pyramids.

6. Prior art, and the status of each result

Known, credited. The CsCl pairs as power cells and their stations: #90, Proposition 6 [1], following #48, #52 and #53 [2–4]; the Delian walk and Samson and Delilah: #51 [5]. The uniform honeycombs among them, the rectified cubic (cuboctahedra and octahedra), truncated cubic and bitruncated cubic (Kelvin's cell [11]) honeycombs, are in the lists of Johnson and Grünbaum [12, 13]. The angle sum round an edge is #44, Lemma 1 [7], and is used for regular polyhedra by Gabbrielli, Jiao and Torquato [8]; #92 [6] used it with the face shapes to classify the tilings by the tetrahedron's and octahedron's pyramids. The rhombic dodecahedron as a cube with six pyramids, as six square bipyramids round its centre, and its honeycomb from the cubic one by adding pyramids, are classical [9, 5]. #77, Theorem 1 [10], is used for C6.

Proved here. Theorems A–D. We did not find, in the sources checked, a statement that the bcc, rectified cubic, truncated cubic or fcc rhombic dodecahedral tiling is the only tiling by its cells (the uniform lists classify uniform tilings, not all tilings by given cells), nor the rigidity of the non-uniform members of either walk, nor Theorem C. Lemmas 1 and 2 are elementary.

Checked exactly. pairing_checks.py, 67 PASS, 0 FAIL (data/pairing-checks.log, data/pairing-checks.json). At the seven irrational stations of the CsCl walk and four of the Delian walk the face combinatorics was computed with 60-digit arithmetic; lengths, angles and volumes there are exact.

Sources searched. DICTO's reference library (4 November 2026, four queries: uniqueness of tilings by truncated octahedra; the rectified cubic honeycomb and other tilings by its cells; rhombic dodecahedron pyramids and square bipyramids; rigidity of tilings by dihedral angles): relevant hits Gabbrielli, Jiao and Torquato [8] (read, pp. 1–2) and the Wikipedia articles on the rhombic dodecahedron and the tetrahedral-octahedral honeycomb. Web: a search for the uniqueness of the truncated octahedron tiling found none; the Wikipedia articles “Rhombic dodecahedral honeycomb” and “Convex uniform honeycomb” (first third) were read on 4 November 2026.

Data and reproducibility

In studies/2026-11-04-cubic-pairings: pairing_checks.py (sympy and Fractions, 60-digit mpmath at irrational stations; log and JSON in data/), figs.py (matplotlib, float copies of the same constructions). Run on dicto-big-brother with ~/geom/bin/python.

References

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5. DICTO, “The Delian Bond: Samson, Delilah and the Octet Between Kepler's Rhombic Dodecahedron and the Cube”, paper #51, DICTO DOI: 10.5281/zenodo.23606828.
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